

MATH3085 Data Analysis Project

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1(a)

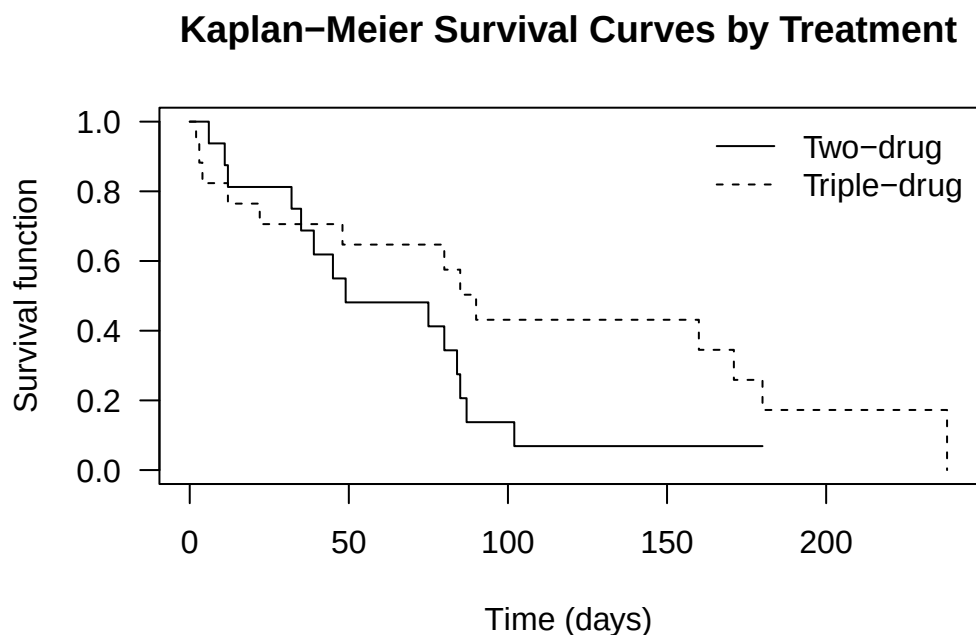


Figure 1: Kaplan–Meier survival curves for the two treatment groups (solid line: two-drug; dashed line: triple-drug).

The Kaplan–Meier survival curves comparing the two-drug and triple-drug treatments are displayed in Figure 1. For the majority of the observed time period, the survival curve for the triple-drug treatment (dashed line) lies above the two-drug curve (solid line). This indicates that patients on the triple-drug regimen had a higher observed probability of remaining relapse-free over time compared to those on the two-drug regimen. Although both groups experienced a similar rate of early relapse within the first 40 days, the triple-drug group showed a more sustained relapse-free period thereafter. This visual pattern suggests a potential benefit from the inclusion of saquinavir, though the small sample size ($n=17$ per group) limits the reliability of these observed patterns, and further statistical analysis is needed to confirm the effect.

1(b)

Table 1: Estimated relapse-free survival probability at 120 days with 95% confidence intervals

Treatment	S.120.	Lower.95..CI	Upper.95..CI
Two-drug	0.069	0.010	0.454
Triple-drug	0.431	0.241	0.771

Although the 95% confidence intervals in the table above overlap, the point estimate for the relapse-free survival probability at 120 days is substantially higher in the triple-drug group (0.431 vs. 0.069), consistent with the separation of the Kaplan-Meier curves observed in Figure 1. The overlap suggests that the evidence for a difference at this specific time point is not conclusive based on the intervals alone.

1(c)

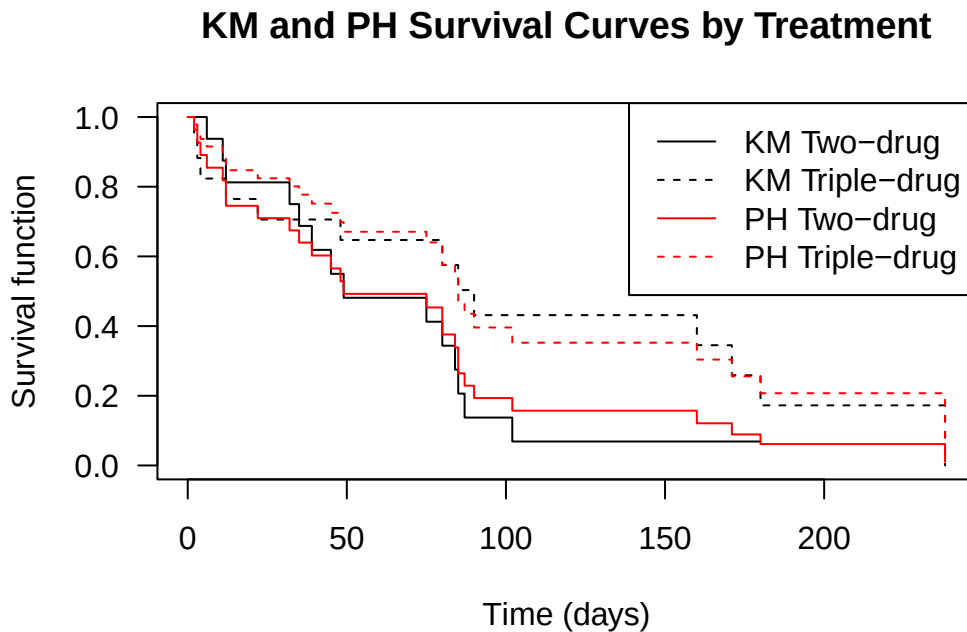


Figure 2: Kaplan-Meier survival curves (black) overlaid with predicted survival curves from the Cox proportional hazards model (red). Solid line: two-drug treatment; dashed line: triple-drug treatment.

The graph overlays the Kaplan-Meier curves (black) with the Cox PH model estimates (red). The PH curves follow the empirical KM trends well, particularly mid-study, supporting the plausibility of the constant hazard ratio assumption. The PH estimates are smoother than the KM step-functions, reflecting the model's semi-parametric structure. Minor deviations at the tails are expected given the small sample size ($n=34$) and the presence of censoring. Overall, the good visual alignment suggests the Cox model

reasonably captures the treatment effect on relapse-free survival. Furthermore, the consistent vertical distance between the two red PH curves visually reinforces the estimated hazard ratio of 0.564, indicating a lower hazard of relapse for the triple-drug regimen compared to the two-drug therapy, although this difference is not statistically significant.

The fitted Cox proportional hazards model is

$$h(t \mid \text{treatment}) = h_0(t) \exp(\beta \cdot I\{\text{treatment} = \text{Triple-drug}\})$$

with the two-drug treatment as the reference group.

The estimated hazard ratio for triple-drug versus two-drug treatment is 0.564 (95% CI [0.255, 1.248]; $p = 0.157$). This indicates that the triple-drug regimen is associated with a 43.6% decrease in the hazard of CD4 relapse at any given time compared to the two-drug regimen (calculated as $(1 - 0.564) \times 100\%$). However, this difference is not statistically significant, indicating that there is insufficient evidence to conclude that the triple-drug treatment reduces the hazard of relapse compared to the two-drug regimen.

1(d)

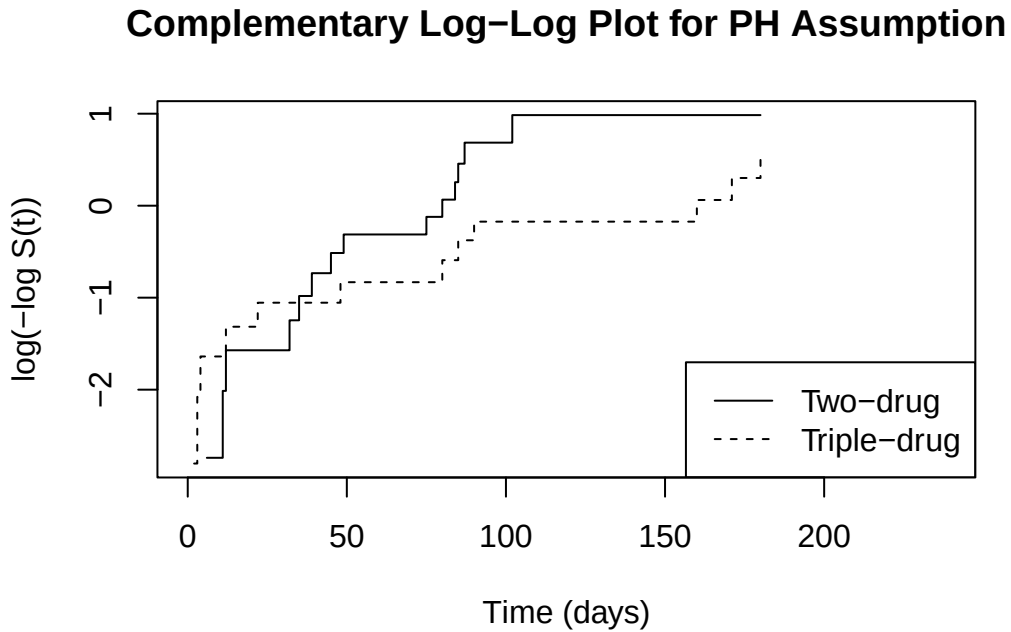


Figure 3: Complementary log-log transformation of Kaplan-Meier survival curves by treatment (solid line: two-drug; dashed line: triple-drug). The curves are approximately parallel, suggesting the proportional hazards assumption is reasonable.

The proportional hazards assumption appears to be reasonable. To assess this, we plot the complementary log-log transformation of the Kaplan-Meier survival estimates, $\log(-\log S(t))$, against time for each treatment group. The resulting curves are approximately parallel, suggesting that the hazard ratio remains constant over time. However, because the sample size is small ($n = 17$ per group), this assessment is limited and minor deviations from proportionality may be present.

2(a)

The data in DRS.csv represent a clinical trial for diabetic retinopathy, with 197 patients randomly assigned to laser treatment or placebo in one eye. The event of interest is blindness (visual acuity < 5/100 at two consecutive tests).

The full Cox proportional hazards (PH) model is

$$h(t | \mathbf{X}) = h_0(t) \exp(\beta_1 \mathbb{I}(\text{treatment} = \text{placebo}) + \beta_2 \text{age} + \beta_3 \text{riskgp}),$$

where $h_0(t)$ is the baseline hazard, **treatment** is a factor with **laser** as the reference level, **age** is continuous (years), and **riskgp** is continuous (values 6 to 12).

To select the preferred model, backward stepwise selection based on AIC was used, starting from the full model.

Age has a high p -value (0.756), indicating no significant effect on the hazard. Backward selection therefore removes age, retaining treatment and riskgp (AIC reduced from 839.7 to 837.8).

The preferred model is

$$h(t | \mathbf{X}) = h_0(t) \exp(\beta_1 \mathbb{I}(\text{treatment} = \text{placebo}) + \beta_3 \text{riskgp}).$$

Table 2: Preferred Model Parameter Estimates

	log(HR)	HR	SE(log(HR))	z	p-value
treatmentplacebo	0.524	1.688	0.217	2.408	0.016
riskgp	0.118	1.125	0.074	1.604	0.109

Table 3: Hazard Ratios with 95% CI

	HR	exp(-coef)	Lower 95%	Upper 95%
treatmentplacebo	1.688	0.592	1.102	2.585
riskgp	1.125	0.889	0.974	1.300

Treatment is statistically significant ($p = 0.016$): patients on placebo have a 69% higher hazard of blindness compared with laser (HR = 1.688, 95% CI [1.102, 2.585]). Risk group has a marginal effect ($p = 0.109$): each one-unit increase in riskgp raises the hazard by 13% (HR = 1.125, 95% CI [0.974, 1.300]). Age was not statistically significant and was removed to improve model parsimony without loss of fit. Overall, laser treatment significantly improves survival (delays blindness), and higher-risk patients tend to have worse outcomes, although the evidence for risk group is weak. The final model adequately captures the dependence of survival on treatment and risk group.

2(b)

Table 4: Efron Approximation

	Coef	HR	SE(Coef)	z	p-value
treatmentplacebo	0.524	1.688	0.217	2.408	0.016
riskgp	0.118	1.125	0.074	1.604	0.109

Table 5: Exact Partial Likelihood

	Coef	HR	SE(Coef)	z	p-value
treatmentplacebo	0.534	1.706	0.221	2.424	0.015
riskgp	0.122	1.129	0.075	1.625	0.104

From the table of event times, there are events with tied survival times. In total, there are 22 distinct tied survival times among 77 distinct events.

From the results table, under the Exact method the estimated coefficients are: Treatment: coefficient = 0.534, SE = 0.221, $p = 0.015$ Risk group: coefficient = 0.122, SE = 0.075, $p = 0.104$

Under the Efron method, the estimates are very similar: Treatment: coefficient = 0.524, SE = 0.217, $p = 0.016$ Risk group: coefficient = 0.118, SE = 0.074, $p = 0.109$

Since the coefficients, standard errors, and p -values from the Exact and Efron methods are very close, there is no evidence of meaningful bias due to the tie-handling method. Therefore, the Efron approximation provides an accurate and reliable alternative to the Exact method for this dataset.

The hazard ratios, standard errors, z-statistics, and p-values are nearly identical between the Efron and exact methods, with only minor differences in precision due to the handling of ties. This suggests that the Efron approximation is sufficient for this data set, and the choice of method does not alter the conclusions.

2(c)

The Weibull AFT model is specified as

$$\log(T) = \beta_0 + \beta_1 \mathbb{I}(\text{treatment} = \text{placebo}) + \beta_2 \text{riskgp} + \sigma\epsilon,$$

where ϵ follows an extreme-value distribution (equivalent to a Weibull distribution for survival times), **laser** is the reference level for treatment, **riskgp** is treated as a continuous covariate, and σ is the scale parameter. plugging in the values below we get $\text{Log}(T) = 6.204 - 0.645(\text{treatmentplacebo}) - 0.144(\text{riskgp}) + 1200 * \epsilon$.

Table 6: Weibull AFT model regression coefficients (β parameters) with 95% confidence intervals.

	Covariate	Estimate	Lower	Upper
treatmentplacebo	treatmentplacebo	-0.645	-1.165	-0.124
riskgp	riskgp	-0.144	-0.317	0.030

The scale parameter is = 1.2

The negative coefficient for treatmentplacebo (-0.645 , 95% CI $[-1.165, -0.124]$) indicates that placebo is associated with shorter survival times compared with laser, by a factor of $\exp(-0.645) \approx 0.525$, suggesting a protective effect of laser against blindness. The negative coefficient for riskgp (-0.144 , 95% CI $[-0.317, 0.030]$) implies that higher risk groups tend to have shorter survival times, consistent with the Cox model, although the confidence interval includes zero, indicating marginal statistical significance.

Log-log Survival Plot to check Weibull Assumption

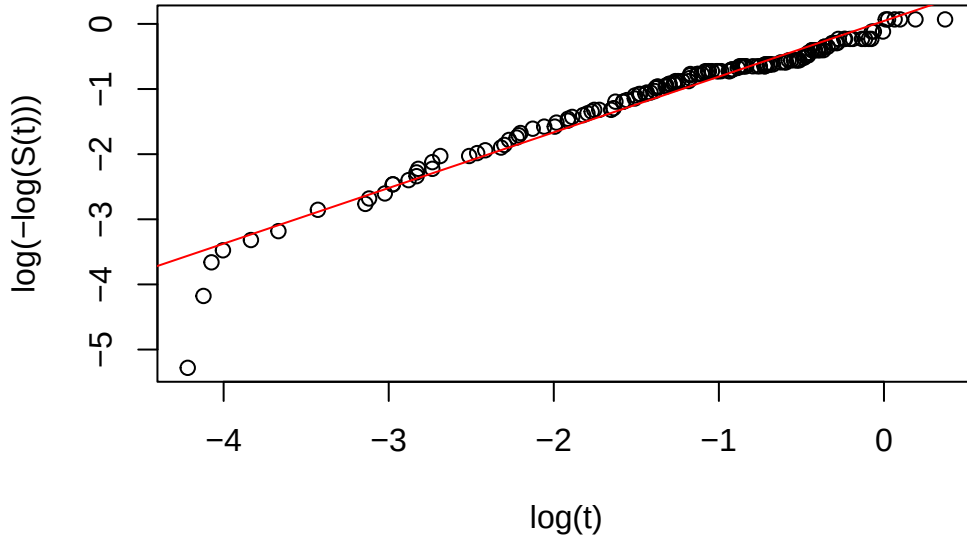


Figure 4: Log-log survival plot ($\ln(-\ln(\hat{S}(t)))$ vs $\ln(t)$). The approximately linear relationship between the log-log survival probability and log-time confirms that the Weibull distribution is an appropriate choice for the baseline hazard in this AFT model.

To assess the suitability of the Weibull distribution for modeling the baseline hazard, a log-log survival plot of $\ln(-\ln(\hat{S}(t)))$ versus $\ln(t)$ was constructed. As shown in Figure 4, the survival curves display an approximately linear trend, suggesting that the Weibull distribution is a reasonable assumption for this dataset. This supports the use of the Weibull accelerated failure time (AFT) model to interpret changes in survival times across different treatment and risk groups. To conclude, I can say this linear pattern confirms the Weibull AFT model provides a reasonable fit to this data.